

guideline for reporting systematic reviews. *BMJ (Clinical Research Ed.)* 2021;**372**:n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]

Riley 2011

Riley RD, Higgins JP, Deeks JJ. Interpretation of random effects meta-analyses. *BMJ (Clinical Research Ed.)* 2011;**342**:d549.

Rizvi 2017

Rizvi S, Ladino LD, Hernandez-Ronquillo L, Téllez-Zenteno JF. Epidemiology of early stages of epilepsy: risk of seizure recurrence after a first seizure. *Seizure* 2017;**49**:46-53. [DOI: [10.1016/j.seizure.2017.02.006](https://doi.org/10.1016/j.seizure.2017.02.006)]

Sander 2003

Sander JW. The natural history of epilepsy in the era of new antiepileptic drugs and surgical treatment. *Epilepsia* 2003;**44** Suppl 1:17-20. [DOI: [10.1046/j.1528-1157.44.s.1.1.x](https://doi.org/10.1046/j.1528-1157.44.s.1.1.x)]

Schünemann 2024

Schünemann HJ, Vist GE, Higgins JP, Santesso N, Deeks JJ, Glasziou P, et al. Chapter 15: Interpreting results and drawing conclusions. In: Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, Welch VA (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.5. Cochrane, 2024.. Available from cochrane.org/handbook.

Snell 2016

Snell KI, Hua H, Debray TP, Ensor J, Look MP, Moons KG, et al. Multivariate meta-analysis of individual participant data helped externally validate the performance and implementation of a prediction model. *Journal of Clinical Epidemiology* 2016;**69**:40-50.

Sterne 2011

Sterne JA, Sutton AJ, Ioannidis JP, Terrin N, Jones DR, Lau J, et al. Recommendations for examining and interpreting funnel plot asymmetry in meta-analyses of randomised controlled trials. *BMJ (Clinical Research Ed.)* 2011;**343**:d4002.

West 2019

West S, Nevitt SJ, Cotton J, Gandhi S, Weston J, Sudan A, et al. Surgery for epilepsy. *Cochrane Database of Systematic Reviews* 2019, Issue 6. Art. No: CD010541. [DOI: [10.1002/14651858.CD010541.pub3](https://doi.org/10.1002/14651858.CD010541.pub3)]

References to other published versions of this review**Adan 2021**

Adan G, Neligan A, Nevitt SJ, Bonnett L, Sander JW, Marson AG. Prognostic factors predicting an unprovoked seizure recurrence in children and adults following a first unprovoked seizure. *Cochrane Database of Systematic Reviews* 2021, Issue 1. Art. No: CD013848. [DOI: [10.1002/14651858.CD013848](https://doi.org/10.1002/14651858.CD013848)]

CHARACTERISTICS OF STUDIES**Characteristics of included studies** [ordered by study ID]**Al-Momani 2020****Study characteristics**

Notes	Jordanian retrospective cohort study Paediatric study: 1 month to 16 years Excluded: previous febrile seizures, absences, myoclonus 290 participants 5-year follow-up 57.9% male sex Funding: no funding information stated
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Retrospective case review
Study attrition	No	Retrospective study, missing data and incomplete data regarding investigations
Prognostic factor measurement	No	Mixed modalities of prognostic factors (imaging - uses CT, MRI and US)
Outcome measurement	Unclear	Incomplete follow-up data

Al-Momani 2020 (Continued)

Adjustment for other prognostic factors	Yes	Univariable and multivariable analysis conducted
Statistical analysis and reporting	Yes	Appropriate statistical methods applied, including modelling of data
Overall risk of bias	Unclear	Moderate

Baldin 2017

Study characteristics

Notes	USA prospective cohort study Adult cohort No mention of ASMs or seizure types 52 participants 2-year follow-up 53.9% male sex Funding: study funded by the Epilepsy Society of Australia
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Item	Authors' judgement	Support for judgement
Study participation	No	Ascertainment methods different for each recruiting site. Poor response rate from those who agreed to take part
Study attrition	Yes	No dropouts reported in the study
Prognostic factor measurement	Unclear	No prognostic factors of seizure recurrence reported
Outcome measurement	Yes	Outcomes are well-defined
Adjustment for other prognostic factors	Yes	Multiple adjustments made for varying factors
Statistical analysis and reporting	Yes	Reasonable statistical methodology applied to data
Overall risk of bias	Unclear	Moderate

Bell 2016

Study characteristics

Notes	British prospective cohort All age cohort 302 participants but missing data in 22 participants (324-22) 23.6 years of follow-up 46% male sex Funding: no specific funding declared. Authors state no financial support was received for this study
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Bell 2016 (Continued)

Item	Authors' judgement	Support for judgement
Study participation	Yes	Prospective well-characterised participants for all categories
Study attrition	Unclear	High dropout rate but prolonged follow-up period mitigates this
Prognostic factor measurement	Yes	Multiple variables measured
Outcome measurement	Yes	Clear definition of outcome and measurement
Adjustment for other prognostic factors	Yes	Multiple adjustments
Statistical analysis and reporting	Yes	Appropriate analysis applied to the data
Overall risk of bias	Yes	Low

Bonnett 2010

Study characteristics

Notes	British multicentre RCT Paediatric and adult ASM given and recurrence reported separately 1443 participants 18 months follow-up 63% male sex Funding: funded by the National Institute for Health Research (NIHR), Health Technology Assessment (HTA) Programme
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Only patients where there was equipose re: starting treatment were included
Study attrition	Unclear	Sizable proportion refused randomisation and another number were lost to follow-up
Prognostic factor measurement	Yes	Multiple risk factors measured
Outcome measurement	Yes	Very clear outcome measurement
Adjustment for other prognostic factors	Yes	Multiple adjustments made
Statistical analysis and reporting	Yes	Excellent stats reporting
Overall risk of bias	Yes	Low

Chen 2016

Study characteristics

Notes	Chinese prospective cohort study Ages of participant: 3 to 77 years No ASMs Excluded: absences, myoclonus and unspecified seizures SE included 134 participants 24-month follow-up 66% male sex Funding: this study was supported by grants from the National Science Council (NSC 99-2314-B-038-017-MY3) and Taipei Medical University (TMU100-AE1-B07)
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Item	Authors' judgement	Support for judgement
Study participation	Yes	Representative sample, consecutive patients, single centre but very clear inclusion criteria
Study attrition	Unclear	Difficult to ascertain patients lost to follow-up from description
Prognostic factor measurement	Yes	Very well-defined and well-documented
Outcome measurement	Yes	Very clear outcomes
Adjustment for other prognostic factors	Unclear	Unclear if controlled for variables that could affect outcomes
Statistical analysis and reporting	Yes	Appropriate analysis performed
Overall risk of bias	Yes	Low

Daoud 2004

Study characteristics

Notes	Jordanian prospective cohort study Paediatric cohort: 3 months to 14 years No ASMs Excluded: absences, myoclonus and infantile spasms SE included 265 participants 3-year follow-up 56.6% male sex Funding: not reported in the manuscript
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Item	Authors' judgement	Support for judgement
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Daoud 2004 (Continued)

Study participation	Yes	Consecutive patients from large teaching hospitals, representative of wider population
Study attrition	Unclear	Unclear re: dropout rates
Prognostic factor measurement	No	All patients did not have uniform investigations, e.g. EEG and brain imaging, due to cost limitations
Outcome measurement	Yes	Detailed reporting of recurrence at several time points that are well described
Adjustment for other prognostic factors	No	No adjustments undertaken
Statistical analysis and reporting	No	Univariable analysis only; no multivariable analysis performed
Overall risk of bias	No	High

Hauser 1990
Study characteristics

Notes	USA prospective cohort All ages included ASMs given (80%) Focal and generalised seizures included SE included 208 participants 22-month follow-up No mention of gender Funding: supported in part by grant NS 12592 from the National Institute of Neurological and Communicative Disorders and Stroke (NINCDS), National Institutes of Health
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Item	Authors' judgement	Support for judgement
Study participation	No	Very narrow inclusion criteria - recruitment had to be within 24 hours
Study attrition	Yes	20 participants out of 208 withdrew from the study
Prognostic factor measurement	Yes	Well described
Outcome measurement	Yes	Well represented
Adjustment for other prognostic factors	Unclear	Some attempt to adjust made
Statistical analysis and reporting	Yes	Appropriate methods used
Overall risk of bias	Unclear	Moderate

Hesdorffer 2007

Study characteristics

Notes	USA retrospective registry. Case-control study All ages included Acute vs unprovoked seizures in stroke, TBI and CNS infections. Stroke (101); TBI (37); CNS infection (10). All seizures including focal aware seizures are included. Only first seizure included if associated with CVA, TBI or CNS infections SE (25.7%) included 148 participants 10-year follow-up No proportional data on gender provided Funding: supported by grant NS 31146 from the National Institute of Neurological Disorders and Stroke (NINDS)
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Item	Authors' judgement	Support for judgement
Study participation	Yes	Representative of wider population
Study attrition	Unclear	Case-control database study
Prognostic factor measurement	Yes	The study clearly defines a range of prognostic factors, applies them consistently, and uses them appropriately in both univariate and multivariate analyses
Outcome measurement	Yes	Well-described and defined outcomes with time points clearly set out
Adjustment for other prognostic factors	Yes	Multiple adjustments made
Statistical analysis and reporting	Yes	Well-described and represented
Overall risk of bias	Yes	Low

Hui 2001

Study characteristics

Notes	Chinese retrospective cohort study. Single centre Age: > 14 years GTCS only No ASMs Excluded: absences, myoclonus, infantile spasms SE excluded 132 participants 27.2 months of follow-up 50% male sex Funding: not reported
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Item	Authors' judgement	Support for judgement
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Hui 2001 (Continued)

Study participation	No	Multiple exclusion criteria makes it a non-representative cohort
Study attrition	Unclear	Retrospective, so attrition is unclear
Prognostic factor measurement	Yes	Multiple prognostic factors
Outcome measurement	Unclear	Seizure recurrence until four years
Adjustment for other prognostic factors	Unclear	Some adjustment undertaken
Statistical analysis and reporting	Yes	Appropriate stats - univariable and multivariable, survival curves included
Overall risk of bias	Unclear	Moderate

Kanemura 2015
Study characteristics

Notes	Japanese prospective cohort Single-centre study Paediatric cohort: 1 month to 15 years No ASMs 87 participants 41% male Minimum 4-year follow-up Eligibility: included individuals with a first unprovoked seizure, including those with prior febrile, neonatal, or post-traumatic seizures Excluded: ambiguous events, acute symptomatic seizures (e.g. with encephalopathy or febrile illness), and cases with structural or progressive brain pathology on MRI as well as SE Funding: not reported
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Retrospective study, unclear how selection occurred
Study attrition	Yes	Nine children who met the criteria did not have follow-up data
Prognostic factor measurement	Yes	Extensive prognostic factors are described in the paper
Outcome measurement	Unclear	Seizure recurrence at four years given
Adjustment for other prognostic factors	No	No adjustments
Statistical analysis and reporting	Unclear	No survival curves
Overall risk of bias	Unclear	Moderate

Machado 2021

Study characteristics

Notes	Brazilian retrospective cohort Paediatric cohort: 1 month to 14 years ASMs given No specific mention of SE but no cases recorded Included: focal and generalised seizures 74 participants 32.5 months follow-up 55% male sex Funding: no funding statement found in the article
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Retrospective study
Study attrition	Unclear	No clearly reported dropout rates
Prognostic factor measurement	Yes	High number of prognostic factors collected and presented
Outcome measurement	Unclear	Recurrence rates presented, but not very clearly defined with time points
Adjustment for other prognostic factors	Unclear	Unclear if adjustments were undertaken
Statistical analysis and reporting	No	No univariable or multivariable modelling undertaken and no survival curves presented
Overall risk of bias	Unclear	Moderate

Mahler 2015

Study characteristics

Notes	Retrospective registry study of first unprovoked seizure in Northern Stockholm, Sweden No ASMs All ages Excluded: seizures occurring < 7 days after a TBI were defined as acute symptomatic and not included, unless followed by at least 1 unprovoked seizure > 7 days after the TBI 1885 cases 55% male sex 6 months of follow-up Funding: supported by a grant from the "Friends of the Tel-Aviv Medical Center" (non-commercial funding source)
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Item	Authors' judgement	Support for judgement
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Mahler 2015 (Continued)

Study participation	No	Clearly defined retrospective study, some conflation of single seizure and multiple seizure presentations
Study attrition	Unclear	N/A - retrospective registry study
Prognostic factor measurement	No	Mostly focussed on a single prognostic factor - TBI, only some mention of other factors
Outcome measurement	Unclear	Appropriately defined outcomes and end points for the study but within a very narrow spectrum of TBI
Adjustment for other prognostic factors	Unclear	Some sensitivity checks for possible confounding factors
Statistical analysis and reporting	Unclear	Appropriate statistical reporting with risk ratios stated clearly from univariable statistics but no multivariable model used in the paper
Overall risk of bias	No	High

Mizorogi 2015
Study characteristics

Notes	Japanese prospective cohort Paediatric cohort No ASMs Excluded: absence, myoclonus seizures Focal and generalised seizures included SE excluded 73 participants 4.6 years of follow-up 45% male sex Funding: no funding statement found in the article
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Clear criteria but some patients excluded e.g. for having organic brain lesions which may not be epileptogenic; also SE excluded
Study attrition	Yes	Less than 10% without follow-up data
Prognostic factor measurement	Unclear	Basic list of factors included
Outcome measurement	Yes	Well documented at various time points
Adjustment for other prognostic factors	Unclear	Unclear if adjustments have been made
Statistical analysis and reporting	Yes	Good and appropriate

Mizorogi 2015 (Continued)

Overall risk of bias	Unclear	Basic list of prognostic factors included. However, it is unclear whether all relevant confounders were measured or appropriately adjusted for in the analysis, and limited detail is provided on the rationale for factor selection.
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Pang 2022
Study characteristics

Notes	Prospective observational study of 239 adults in Australia who had a first unprovoked seizure from sleep only between 2000 and 2011. Patients were reviewed at clinic 3 to 9 months after the index seizure. If there were ongoing seizures, patients were followed up in the epilepsy clinic. If there were no further seizures, patients were discharged and contacted by phone every 1 to 2 years until seizure recurrence or death. ASM given 61% male sex Funding: no funding information provided
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Item	Authors' judgement	Support for judgement
Study participation	No	Only sleep onset unprovoked seizure
Study attrition	Unclear	No mention of study dropout rates
Prognostic factor measurement	Yes	High number of prognostic factors measured and provided in easily accessible format
Outcome measurement	Yes	Clear outcomes and endpoints
Adjustment for other prognostic factors	Yes	Significant variables controlled for in multivariable model
Statistical analysis and reporting	Yes	Appropriate use of statistical analysis and clear reporting of univariable and multivariable statistics
Overall risk of bias	Unclear	Moderate

Qadri 2017
Study characteristics

Notes	Observational prospective study from March 2016 to December 2016 Infants 1 to 12 months presenting to the hospital with first-time seizure Srinagar, India in a tertiary care hospital for children No ASMs given Excluded: infants with any significant perinatal history or with prior history of seizures 50 infants included 58% male sex Funding: no funding information provided
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Qadri 2017 (Continued)

Item	Authors' judgement	Support for judgement
Study participation	Unclear	Consecutive children with first unprovoked seizure recruited prospectively
Study attrition	Yes	Six of 56 infants lost to follow-up and clearly stated
Prognostic factor measurement	No	Very limited prognostic factors measured and reported in the paper
Outcome measurement	Unclear	Clear outcomes and endpoints but short follow-up period
Adjustment for other prognostic factors	Unclear	No mention of confounding factors
Statistical analysis and reporting	No	Only univariable statistics with no mention of a multivariable model
Overall risk of bias	No	High

Ramos Lizana 2000

Study characteristics

Notes

Prospective cohort study
Patients aged less than 14 years with 1 or more unprovoked seizures
Spain
Minimum 6 months of follow-up; average follow-up 3.8 years
ASMs given
Excluded: neonatal convulsions and patients already receiving antiepileptic treatment or who had been examined at other centres
217 patients
61% male sex
Funding: no funding information provided

Item	Authors' judgement	Support for judgement
Study participation	Unclear	Consecutive patients under the age of 14 included; unclear why this cut off was used for a paediatric cohort
Study attrition	Yes	Only three children lost to follow-up out of 217
Prognostic factor measurement	Yes	High number of prognostic factors measured and clearly reported
Outcome measurement	Yes	Clear endpoints defined and reported
Adjustment for other prognostic factors	Unclear	No mention of confounding factors
Statistical analysis and reporting	Yes	Appropriate univariable and multivariable analysis
Overall risk of bias	Yes	Low

Ramos Lizana 2009

Study characteristics

Notes	Prospective cohort study Patients aged 14 years or younger who attended for a first remote symptomatic unprovoked seizure Spain Mean follow-up: 4.1 years ASM given Excluded: neonatal convulsions, patients already on antiepileptic treatment, those who had been examined previously in other centres, and those whose detailed clinical record revealed antecedents of prior seizures 63 patients 56% male sex Funding: no funding information provided
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Consecutive patients aged 14 years or younger who attended the unit for a first remote symptomatic unprovoked seizure
Study attrition	Yes	Only one patient lost to follow-up of 63 participants included
Prognostic factor measurement	Yes	Adequate reporting of prognostic factors and a high number of factors measured
Outcome measurement	Yes	Clear study end points defined and reported
Adjustment for other prognostic factors	Unclear	No formal discussion of confounding factors but multivariable model used
Statistical analysis and reporting	Yes	Appropriate statistical analysis and reporting with univariable and multivariable statistics clearly stated
Overall risk of bias	Yes	Low

Sathirapanya 2020

Study characteristics

Notes	Retrospective case note review Patients aged ≥ 15 years who presented to the emergency department with a first seizure Thai study Median follow-up 33.4 months ASM given Excluded: those with a known history of epilepsy and all epilepsy mimics 414 cases 64.5% male sex Funding: no funding information provided
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Item	Authors' judgement	Support for judgement
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Sathirapanya 2020 (Continued)

Study participation	No	Retrospective case review of patients ≥ 15 year but only those who were admitted for at least 48 hours
Study attrition	Unclear	N/A - retrospective study
Prognostic factor measurement	Yes	High number of prognostic factors that are clearly defined and reported
Outcome measurement	Unclear	Endpoints of study clearly defined and reported
Adjustment for other prognostic factors	Unclear	No formal mention of study confounders
Statistical analysis and reporting	Yes	Appropriate use of univariable and multivariable statistics
Overall risk of bias	Unclear	Retrospective design with selection limited to hospitalised patients staying ≥ 48 hours raises concerns about representativeness. Prognostic factors were reported but with limited detail. Other domains were acceptable, so overall risk of bias is rated as unclear.

Schreiner 2003

Study characteristics

Notes	German prospective cohort Adult cohort > 16 years No ASMs given Excluded: absences and myoclonus Included: all seizure types including SE 157 participants 34 months of follow-up 62% male sex Funding: no funding information provided
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Adults coming to emergency dept with unprovoked seizure, clear criteria for study however only those with an EEG within 48 hours included, which is not representative of the wider population
Study attrition	Yes	Seven patients lost to follow-up
Prognostic factor measurement	Unclear	Extensive for EEG measures but few other factors considered e.g. imaging
Outcome measurement	Yes	Well presented
Adjustment for other prognostic factors	Unclear	Unclear adjustments made
Statistical analysis and reporting	Yes	Survival curves and multivariable/univariable analysis performed

Schreiner 2003 (Continued)

Overall risk of bias	Unclear	Moderate
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Scotoni 2004

Study characteristics

Notes	Brazilian prospective cohort study Paediatric cohort: 1 month to 17 years No ASMs Excluded: absence, myoclonic seizures, infantile spasms, and symptomatic seizures Included: focal and generalised seizures SE (6%) included 213 participants 25.7-month follow-up period 58% male sex Funding: no funding information provided
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Item	Authors' judgement	Support for judgement
Study participation	Unclear	Clear criteria but excluded absence and myoclonus which will introduce some bias
Study attrition	Unclear	16% of participants had less than six months of follow-up
Prognostic factor measurement	Yes	Extensive
Outcome measurement	Yes	Well reported
Adjustment for other prognostic factors	Yes	Plenty of adjustments
Statistical analysis and reporting	Yes	Appropriate survival curves, univariable and multivariable analyses
Overall risk of bias	Yes	Low

Shinnar 2000

Study characteristics

Notes	USA prospective cohort Age: 1 month to 19 years No ASMs All seizures included No specific mention of SE 407 participants 9.6 years of follow-up No mention of gender Funding: this work was supported by the National Institutes of Health (NIH) grant NS 31146 from the National Institute of Neurological Disorders and Stroke (NINDS)
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Shinnar 2000 (Continued)

Item	Authors' judgement	Support for judgement
Study participation	Unclear	Clear criteria but excluded absence and myoclonus which will introduce some bias
Study attrition	Unclear	16% of participants had less than six months of follow-up
Prognostic factor measurement	Yes	Extensive
Outcome measurement	Yes	Well reported
Adjustment for other prognostic factors	Yes	Plenty of adjustments
Statistical analysis and reporting	Yes	Appropriate survival curves, univariable and multivariable analyses
Overall risk of bias	Yes	Low

Takami 2021

Study characteristics

Notes	Japanese prospective observational study 467 participants Patients aged between 1 month and 16 years who were attended to for their first unprovoked seizures and had an EEG at the Japanese Red Cross Society Himeji Hospital between 1 November 2008 and 31 October 2016 No ASMs 56.5% male sex Funding: this study was supported by a grant from the Practical Research Project for Rare/Intractable Diseases from the Japan Agency for Medical Research and Development (AMED) (JP20ek0109486)
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Item	Authors' judgement	Support for judgement
Study participation	No	Only children who had an EEG were included in the study
Study attrition	Unclear	No mention of study attrition or dropouts from study
Prognostic factor measurement	Yes	High number of prognostic factors collected as part of study
Outcome measurement	Yes	Clear study end points and outcomes clearly defined
Adjustment for other prognostic factors	Unclear	No formal reporting of study confounders but they do use multivariable models to control for a number of variables
Statistical analysis and reporting	Yes	Appropriate use of univariable and multivariable statistics

Takami 2021 (Continued)

Overall risk of bias	Unclear	Moderate
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Winckler 2004
Study characteristics

Notes	Brazilian prospective cohort study Paediatric cohort: 1 month to 16 years No ASMs All seizure types included No specific mention of SE 109 participants 24 months of follow-up 64.2% male sex Funding: no funding sources mentioned or declared
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Item	Authors' judgement	Support for judgement
Study participation	Yes	Consecutive patients, really clear inclusion and exclusion criteria
Study attrition	Unclear	Not clearly stated
Prognostic factor measurement	Yes	Very extensive collection of factors included
Outcome measurement	Yes	Very clear outcome and risk for epilepsy development given at multiple time points
Adjustment for other prognostic factors	Yes	Multiple adjustments
Statistical analysis and reporting	Yes	Well-reported methods
Overall risk of bias	Yes	Low

ASMs: anti-seizure medications
 CNS: central nervous system
 CT: computed tomography
 CVA: cerebrovascular accident
 GTCS: generalised tonic-clonic seizure
 MRI: magnetic resonance imaging
 RCT: randomised controlled trial
 SE: status epilepticus
 TBI: traumatic brain injury
 US: ultrasound

Characteristics of excluded studies [ordered by study ID]